Khadijah Aamir Chaudhry

Pakistan

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EDUCATION

Lahore University of Management Sciences *Sept 2023 – May 2027*

BSc Computer Science

Lahore Grammar School Phase 1 DHA, 1-A-1 *May 2023 – May 2021*

A-Level – 4A, O-Level – 9A 1B*

ACADEMIC PROJECTS

Mobile Eye Prescription Estimation using ToF Sensors *Jan 2026 – May 2026*

- Conducted a data camp to ethically gather eye data from 100+ volunteers in underrepresented communities lacking healthcare access
- Designed a smartphone-based objective refraction system using infrared Time-of-Flight (ToF) sensors
- Built a pipeline to extract and reconstruct ocular features from depth and IR data, overcoming limitations of standard depth maps
- Applied physics-based and unsupervised deep learning approaches for robust feature extraction across noisy sensor inputs.

ESP32 Evaluation of Post-Quantum Cryptographic Algorithms *Sept 2025 – Dec 2025*

- Implemented and benchmarked Kyber-512, Light-SABER, and Dilithium-II on an ESP32 microcontroller
- Measured latency, memory usage, and computational overhead under constrained hardware conditions
- Evaluated the feasibility of post-quantum cryptography at the edge for scalable secure systems

Spotify Mood Classification & Recommendation System *Jan 2025 – May 2025*

- Built ML pipeline on a real-world music dataset for mood classification and clustering
- Applied PCA for dimensionality reduction and K-Means for unsupervised grouping of songs
- Performed data cleaning, feature engineering, and model evaluation on noisy, real-world data

PROFESSIONAL EXPERIENCE

Uplift AI – *Data Collection Coach* *July 2024 – Aug 2024*

- Trained and led a team of 10 individuals to ethically and accurately collect audio data for a Pakistani language AI model.
- Held daily quality review sessions, rated submissions, and provided feedback to improve collection standards.
- Organized a field trip to diversify voice data sources and ensure cultural relevance.

LEADERSHIP AND CO-CURRICULARS

LUMS SPADES – *Women in STEM Coordinator* *Sept 2024 – May 2025*

- Led the Women in STEM department in organizing bonding events, digital and financial literacy seminars, and launched PSIFI's first women-only STEM event with over 100 participants.

Umeed-e-Nau NGO – *Co-Founder and Director Logistics* *Sept 2021 – May 2023*

- Co-founded an NGO focused on education and character development for underprivileged and disabled children.
- Directed logistics for lessons, planned tailored activities, and ensured special accommodations.
- Coordinated visits to Rising Sun, Children's Foundation, SOS School, and more.

HONOURS & AWARDS

Merit Scholarship (100%) – Lahore Grammar School Phase 1 DHA

Shortlisted Writing Submission – LUMSx Nexus

Honorable Mention – EducationUSA Pakistan Writing Contest

SKILLS & INTERESTS

Skills: C++, Python, JavaScript, Embedded system design, Machine learning and data science, User research methods

Interests: Literature and Bibliotherapy, Policy and Digital Governance, Tennis, Women in STEM